Optimizing Active Inference Agents for Robustness to Adversarial Behavior in Traffic Interactions

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***Abstract*—Active inference models of traffic interaction assume that explicit communication, such as turn signals, is broadly reliable. Imagine instead a driver who flips on their yield indicator while continuing to accelerate into the intersection. Such mismatches between communicated intent and observed motion can mislead an autonomous agent into yielding when it should not, or holding when it should brake. In a simulated two-agent intersection, a naive trusting agent collides in 100% of deterministic adversarial encounters, where adding a simple inconsistency detector with a threshold τ eliminates the failure. We formulate the selection of τ as an optimization problem in three ways: a scalarized weighted-sum loss solved by golden-section search; a constrained minimization of latency subject to a false alarm bound, solved by both quadratic-penalty and log-barrier methods; and a multi-objective Pareto analysis. The optimal τ varies by $20\times$ across loss weights and $25\times$ across adversarial-prior assumptions. Compared to a ground-truth oracle, the detection-enhanced agent recovers 100% of preventable collisions at the studied noise level. The 'right' threshold is a deployment decision, not a universal constant.**

I. Introduction

Picture a four-way intersection with no traffic signal. Two drivers approach on crossing paths at nearly the same moment—close enough that formal right-of-way rules don't resolve the ambiguity—and they must coordinate by reading each other's intent. One driver activates their turn signal (the conventional gesture for "I am yielding") but keeps their foot on the accelerator. The other driver, seeing the signal, eases off the brake and proceeds into the intersection. The result is a collision that occurs precisely because the trusting driver believed the signal over the motion.

Autonomous driving systems face the same problem, and recent active inference models of traffic interaction [1] treat disagreements between signal and motion as opportunities for uncertainty reduction, with agents minimizing expected free energy (EFE) while inferring the intent of others. These models, however, assume that explicit communication is broadly reliable. We focus on the case where it is *not*: an adversary that signals an intent to yield while behaving inconsistently with that signal.

The mismatch between communicated intent and observed kinematics admits a natural detection rule – maintain a running inconsistency score over the interaction and flag adversarial behavior when the score exceeds some threshold τ. The score and threshold together control the well-known tension between detection latency, false alarm rate, and missed-detection rate. Setting τ too low produces false alarms on honest agents behaving noisily; set it too high, and adversaries are missed until it is too late to brake. This paper studies τ as the design variable of an optimization problem, formulated three ways: (1) as a scalarized minimization of a weighted loss, (2) as a constrained minimization of latency subject to a false alarm bound, (3) and as a multi-objective Pareto analysis over latency, false alarms, and collision rate.

The optimization problem has substance of its own. The objective is stochastic (each evaluation runs an ensemble of randomized episodes), gradient-free (the simulator is not differentiable), and unimodal but with regions of plateau (where the inconsistency score saturates against an adversary or stays at the noise floor against honest agents). The constrained version has the unconstrained latency minimum on the infeasible side of the constraint, so the constraint binds at the optimum– which is the natural setting for both quadratic-penalty and log-barrier methods. We find that all three formulations produce meaningfully different optima depending on weight choices or constraint values, that explicit constraint formulation outperforms weighted-sum proxies when a hard bound exists, and that a sensitivity analysis over observation noise identifies the regimes in which threshold optimization is a meaningful problem at all.

Contributions

- A simulated two-agent intersection environment instrumented for adversarial signaling, in which a naive active-inference agent collides on every encounter and a detection-equipped agent matches a ground-truth oracle in collision prevention.
- Three optimization formulations of the threshold-selection problem (scalarized, constrained, and multi-objective) solved respectively with golden-section search, quadratic-penalty and log-barrier outer loops, and a 50-point Pareto sweep.
- Empirical comparison showing that five of six F1/F2 optima land on the Pareto frontier, that penalty achieves lower latency than weighted-sum proxies at the same false alarm bound, and that the optimal threshold varies by $20\times$ across loss weights and $25\times$ across adversarial-prior assumptions.

II. Related Work

Three threads of prior work touch the problem we study: active inference models of multi-agent interaction, adversarial robustness in autonomous systems, and gradient-free constrained optimization. Each addresses a piece of what we do, but none frames detection-threshold selection in active inference agents as an optimization problem with explicit measurable trade-offs.

a) Active inference for traffic interaction: Active inference [2] formalizes perception and action as joint minimization of expected free energy (EFE) where agents select actions that both reduce uncertainty about latent states and achieve preferred outcomes. Recent work has applied this framework to road-user interactions, modeling space-sharing conflicts as Bayesian inference over the other agent's intent under both kinematic and explicit-signal observations [1]. That work establishes the framework and shows that misleading communication *can* produce unsafe outcomes, but treats this finding as a limitation to be acknowledged rather than addressed. We instead make that failure mode the explicit subject of an optimization problem, asking what threshold value (and what optimization formulation) best mitigates it. We also adopt a simplified EFE objective with the entropy term omitted as passive observation in our environment makes it nearly invariant to action. A richer setting with active signaling would re-engage the full canonical formulation.

b) Adversarial robustness in multi-agent coordination: Robustness against malicious or anomalous behavior of other agents is active in autonomous driving more broadly, and the dominant approach is predictive: forecast the other agent's trajectory under possible behavioral hypotheses, plan against the most adversarial, and hope the forecast is accurate enough. Recent work on intersection navigation has used deep reinforcement learning to learn policies under uncertainty about other drivers' intent [3], while work on adversarial robustness has shown that even small, physically realizable perturbations to other agents' trajectories can substantially degrade trajectory prediction [4]. Both lines treat the other agent as a black-box mover and largely ignore explicit communication channels. The closest classical analog to our setting is the family of 'cheap talk' games [9], in which agents exchange unverifiable messages and the recipient must decide how much to update on them. Cheap talk analysis, however, is qualitative and characterizes when communication is informative in equilibrium. Our contribution is the corresponding quantitative question: given an explicit detection mechanism, what threshold optimally trades latency, false alarms, and missed detections?

c) Gradient-free and constrained optimization: Our methodology rests on standard tools, and the novelty lies in their combination rather than in any individual method. The objective is stochastic and non-differentiable, so we rely on gradient-free methods. Here, golden-section search is the textbook approach for 1D unimodal minimization on a bounded interval [5]. For the constrained formulation we apply two classical methods: the quadratic penalty method [5], [6], which solves a sequence of unconstrained problems with a growing penalty on constraint violation, and the logarithmic barrier method [5], [7], which restricts iterates to the strict interior of the feasible set. Both methods have known characteristic behaviors near the optimum—penalty approaching from the infeasible side, barrier from the strictly feasible side—and we observe these signatures directly in our experiments. For the multi-objective view, we use the standard Pareto frontier construction [5], [8], which serves a dual role: a third formulation and an empirical validation tool for the first two. To our knowledge, this is the first treatment of detection-threshold selection in active inference agents as an explicit optimization problem with cross-formulation empirical comparison.

III. System Model

We consider a two-agent uncontrolled intersection in which agents A and B approach on perpendicular paths. Each agent's state at time t is the pair (d_t, v_t), where d_t is its scalar distance to the intersection along its own axis and $v_t \geq 0$ is its velocity. The intersection is a square box of side W centered at the origin. A collision occurs if and only if both agents are inside the box simultaneously. Each agent additionally emits a binary yielding signal $\sigma_t \in \{0, 1\}$ communicating an intent to yield.

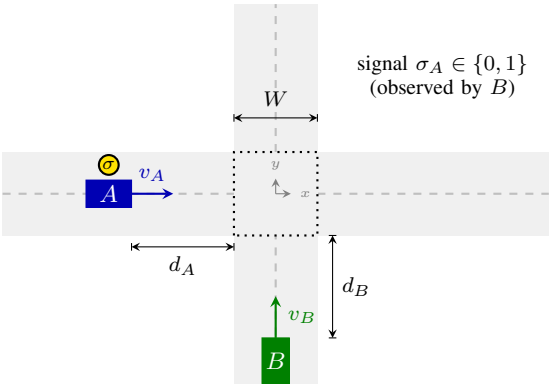


Fig. 1. System schematic. Agents A and B approach a perpendicular intersection along their respective axes with velocities v_A, v_B and distances-to-intersection d_A, d_B. The intersection is a box of side W; a collision occurs iff both agents occupy the box at the same step. Each agent emits a binary yielding signal σ_t observed by the other. In our adversarial setting, A may emit $\sigma_t = 1$ (claiming yield) while maintaining velocity.

A. Active Inference Agent

Agent B, the protagonist, must decide at each timestep whether to proceed or yield without certainty about what the other agent will do. We model this uncertainty explicitly as B maintains a scalar belief $p_t \in [0, 1]$ that the other agent will yield. Using Bayes' rule, the belief is then updated each step from two pieces of evidence: (1) the other agent's observed acceleration and (2) any yielding signal it emits. Concretely, p_t is the posterior probability of a hidden binary type $\theta \in \{\mathrm{Y}, \mathrm{N}\}$ (Yielder or Non-yielder) that is fixed for the episode. Observations are $o_t = (a_t, \sigma_t)$, where a_t is B's estimate of A's acceleration, corrupted by Gaussian noise with standard deviation σ_{obs}.

Conditional on θ, the likelihood factorizes as

$$P(o_t \mid \theta) \;=\; \mathcal{N}(a_t; \mu_\theta, \sigma_\ell^2)\,\mathrm{Bern}(\sigma_t; q_\theta), \tag{1}$$

with parameters $\mu_{\mathrm{Y}} < 0$ (yielders decelerate), $\mu_{\mathrm{N}} \geq 0$ (non-yielders do not), $q_{\mathrm{Y}} \approx 1$ (yielders almost always signal yield), and $q_{\mathrm{N}} \approx 0$ (non-yielders almost never do). For numerical stability, the Bayesian update is performed in log-odds form.

Given a current belief, the agent selects an action by minimizing the simplified EFE objective. Intuitively, the objective trades off 'the risk of colliding if I take this action given my belief about the other agent' against 'the time I expect to spend in the conflict.' Formally, over a discrete action set $\mathcal{A} = \{-a_{\max}, 0, +a_{\max}\}$:

$$J(a) \;=\; \alpha\, P_{\mathrm{coll}}(a, p_t) \;+\; \beta\, T_{\mathrm{cross}}(a), \tag{2}$$

where $P_{\mathrm{coll}}(a, p_t)$ is the predicted collision probability under action a. This is computed by rolling out both agents' trajectories under candidate a and a constant-velocity assumption for the other agent, with a soft safety margin on the predicted temporal gap between intersection-occupancy intervals. $T_{\mathrm{cross}}(a)$ is the predicted time to clear the intersection. We omit the entropy term of the canonical EFE formulation because, with passive observation, $H(p_{t+1}|a)$ varies negligibly across actions in this environment.

B. Adversarial Behavior

Adversarial agents emit $\sigma_t = 1$ (claiming yield) while maintaining their velocity, i.e. a contradiction between explicit communication and observed kinematics. To prevent the detection problem from collapsing onto a known onset time, the adversarial signal begins at a randomized step $t_0 \in [0, 30]$.

C. Inconsistency Detection

We define a per-step inconsistency raw score that is high when the other agent signals yield while moving fast *and* failing to decelerate:

$$r_t \;=\; \begin{cases} \max\bigl(0,\, a_t - a_{\mathrm{yield}}\bigr) & \text{if } \sigma_t = 1 \text{ and } v_t > v_{\min}, \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

where a_{yield} is a yielding-acceleration threshold and $v_{\min}$ gates the score so that honest yielders coasting at low velocity do not contribute. The raw score is smoothed by a leaky exponential average:

$$s_t \;=\; (1 - \lambda)\, s_{t-1} \;+\; \lambda\, r_t. \tag{4}$$

Detection fires the first time $s_t > \tau$ and the trigger is sticky for the remainder of the episode. Once triggered, the agent's belief update zeroes the signal term s.t. kinematic evidence alone drives p_t. The score is gated to the conflict phase (only evaluated while the other agent has not fully cleared the intersection) to avoid false alarms during legitimate post-yielding acceleration.

IV. Optimization Formulations

The single design variable of this threshold-selection problem is τ, the detection threshold, of which we study three different formulations.

A. F1: Scalarized Minimization

The simplest formulation collapses the three metrics we care about—how quickly the detector fires, how often it false-alarms, and how often it fails to fire at all—into a single weighted scalar. The weights encode the deployer's preferences. We then search for the threshold τ that minimizes this scalar:

$$L(\tau) \;=\; w_\ell\, \ell(\tau) \;+\; w_f\, \mathrm{FA}(\tau) \;+\; w_m\, \bigl(1 - \rho(\tau)\bigr), \tag{5}$$

where $\ell(\tau)$ is mean detection latency (in seconds, replaced by a default of 5 s on missed detections so L is well-defined as $\tau \to \infty$), $\mathrm{FA}(\tau)$ is the false alarm rate, and $\rho(\tau)$ is detection recall on adversarial episodes. We solve $\min_\tau L(\tau)$ over $\tau \in [0.01, 1.0]$ by golden-section search on the bounded interval. To probe sensitivity, we report optima at four weight settings: balanced $(w_\ell, w_f, w_m) = (1, 5, 5)$, latency-heavy $(5, 1, 1)$, FA-heavy $(1, 10, 1)$, and miss-heavy $(1, 1, 10)$.

B. F2: Constrained Minimization

A weighted sum forces the deployer to commit numerical preferences to all three metrics jointly. Often what one actually wants is harder to express that way — "minimize latency, but never let the false alarm rate exceed 5%." We pose this directly as a constrained problem:

$$\min_{\tau}\; \ell(\tau) \quad \text{s.t.} \quad \mathrm{FA}(\tau) \leq \mathrm{FA}_{\max}, \tag{6}$$

with $\mathrm{FA}_{\max} = 0.05$. The unconstrained latency minimum sits at low τ (aggressive detection) and the constraint binds for low-τ solutions, making the problem non-trivial.

We solve (6) using two outer-loop methods, each calling golden-section search as the inner 1D optimizer. An *outer iteration* solves one inner unconstrained subproblem at a fixed value of a schedule parameter. The parameter then advances, toward stronger penalty enforcement or weaker barrier strength, and the inner solve is repeated. The two methods differ in how the constraint is enforced:

- *Quadratic penalty:*
  minimize $\ell(\tau) + \rho \cdot \max(0, \mathrm{FA}(\tau) - \mathrm{FA}_{\max})^2$ over a geometric schedule $\rho \in \{1, 10, 100, 1000\}$. The penalty grows with each outer iteration, eventually overwhelming any incentive to violate the constraint.

- *Logarithmic barrier:*
  minimize $\ell(\tau) - \mu \log\bigl(\mathrm{FA}_{\max} - \mathrm{FA}(\tau)\bigr)$ over $\mu \in \{1, 0.1, 0.01, 0.001\}$. The barrier diverges at the constraint boundary, keeping iterates strictly feasible. As μ shrinks, the barrier weakens and the iterate approaches the boundary. The barrier method requires a feasible starting bracket, which we obtain by a coarse linear pre-sweep.

C. F3: Multi-Objective Pareto Analysis

Finally, we treat the trade-off explicitly as a three-objective problem—minimizing $\ell(\tau)$, $\mathrm{FA}(\tau)$, and the collision rate $c(\tau)$—and recover the Pareto frontier by evaluating τ over a 50-point uniform grid and marking non-dominated points.

Overlaying the optima from F1 and F2 on this frontier allows empirical validation; a correctly specified scalar or constrained problem should land on or near a Pareto-optimal point.

D. Evaluation Harness

A single evaluation of any objective at a given τ runs $N = 200$ episodes, seeded deterministically from a base seed. Each episode samples initial distances $d_0 \in [15, 30]$ m and velocities $v_0 \in [8, 12]$ m/s. Adversarial episodes occur at a configurable rate (default 0.5). Honest episodes draw A's policy uniformly from a yielder and a non-yielder. Deterministic seeding makes $L(\tau)$ a deterministic function of τ, which is essential for gradient-free optimization on a stochastic underlying system.

V. Results

We report results in three parts: (*A*) a baseline failure mode that motivates the detection problem, (*B*) comparison of the three optimization formulations against the empirical Pareto frontier, and (*C*) sensitivity analyses over observation noise, the adversarial-episode prior, and an oracle detector that bounds achievable performance.

A. Baseline Failure and Fix

We first establish that the adversarial scenario is non-trivial. In a deterministic head-on test case with the deceptive adversary emitting $\sigma_t = 1$ from $t = 0$, while maintaining velocity, an EFE agent that naively trusts signals has its belief saturate at $p_t \to 1$ within 9 timesteps and collides every time. Adding the detector with $\tau = 0.3$ triggers within 0.2 s of deception onset, after which the belief is driven only by kinematic evidence and falls to near zero, allowing the agent to yield successfully.

B. Pareto Frontier and Formulation Comparison

A 50-point sweep over $\tau \in [0.02, 1.0]$ at noise level $\sigma_{\text{obs}} = 2.0\,\text{m/s}^2$ and adversarial rate 0.5 produces a frontier of 19 non-dominated points (Fig. 3). The frontier traces a smooth monotone curve from low-FA / high-latency (FA $\approx 3\%, \ell \approx 0.34\,\text{s}$) to high-FA / low-latency (FA $\approx 17.5\%, \ell \approx 0.08\,\text{s}$), demonstrating a genuine, continuous trade-off rather than a degenerate two-point landscape.

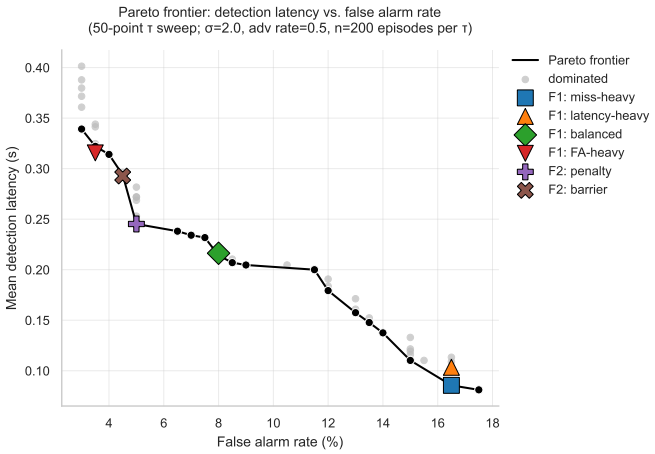


Fig. 3. Pareto frontier in (false alarm rate, detection latency) space, with F1 and F2 optima overlaid. Five of the six optima land on the empirical frontier; F1-balanced is dominated by $\tilde{9}$ ms of latency at the same FA rate.

The four F1 optima are well-separated along the frontier (Table I) from miss-heavy weights, which select $\tau^* = 0.04$ (high recall, high FA), through balanced weights at $\tau^* = 0.54$, to FA-heavy weights at $\tau^* = 0.81$—a $20\times$ range in τ driven entirely by the relative emphasis among the three loss terms. Five of the six F1/F2 optima land directly on the empirical Pareto frontier. F1-balanced is dominated by approximately 9 ms of latency at the same FA rate, a known artifact of weighted-sum scalarization producing non-Pareto solutions when weights happen to align with non-extreme points.

The two F2 methods bracket the constraint boundary characteristically (Fig. 2). The quadratic penalty converges to FA = 5.00% (exactly at the bound, allowing pre-feasible violation during early iterations with small ρ), while the log barrier converges to FA = 4.50% (strictly feasible throughout, approaching the boundary as $\mu \to 0$). Crucially, F2-penalty achieves $\ell = 0.245\,\text{s}$ versus F1-FA-heavy's $\ell = 0.316\,\text{s}$ at comparable FA—an empirical confirmation that explicit constraint formulation outperforms weighted-sum proxies when the practitioner cares about a hard FA bound.

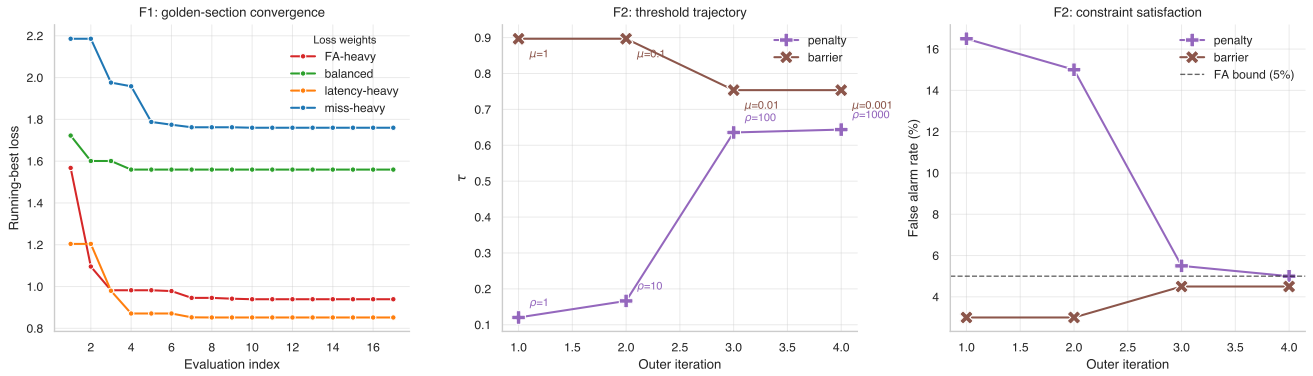


Fig. 2. Convergence of Formulations 1 and 2. Left: running-best loss vs. evaluation index for the four F1 weight settings under golden-section search. Center: τ trajectory for F2 penalty and barrier. Right: false alarm rate vs. outer iteration with the 5% bound marked.

TABLE I
OPTIMA FROM EACH FORMULATION AT $\sigma_{\text{obs}} = 2.0$, REPORTED AS MEAN $\pm$ STANDARD DEVIATION ACROSS 10 RANDOM SEEDS.

Method	τ^*	ℓ (s)	FA (%)	Recall (%)
F1: miss-heavy	0.041±0.000	0.076±0.001	15.55±0.16	84.17±0.12
F1: latency-heavy	0.041±0.000	0.076±0.001	15.55±0.16	84.17±0.12
F1: balanced	0.536±0.001	0.206±0.001	6.50±0.00	81.38±0.15
F1: FA-heavy	0.809±0.000	0.321±0.003	2.50±0.00	73.18±0.33
F2: penalty	0.630±0.000	0.238±0.002	5.00±0.00	79.70±0.33
F2: barrier	0.716±0.000	0.277±0.002	4.00±0.00	77.84±0.33

C. Sensitivity Analyses

The preceding results were reported at a single noise level and adversarial rate. Here, we vary each in turn to identify the regimes where threshold optimization is a meaningful design problem, benchmark the detector against a ground-truth oracle to bound achievable performance, and confirm that the reported optima are stable across random seeds. Together, these analyses characterize when our framework matters and how close it gets to the best possible outcome.

a) Observation noise: At low noise ($\sigma_{\text{obs}} \leq 1.0$), the F1 (balanced) optimum is at the lower boundary of the search interval ($\tau^* \approx 0.02$), reflecting that without observation noise the detector's score plateau against a deceptive adversary is well-separated from the noise-floor against honest agents and no real trade-off exists. The trade-off emerges at $\sigma_{\text{obs}} = 1.5$, where the optimum moves to $\tau^* = 0.35$, and is fully developed at $\sigma_{\text{obs}} = 3.0$ with $\tau^* = 0.61$ (Fig. 4, left). This characterizes precisely the regime in which threshold optimization is a meaningful design problem rather than a degenerate one.

b) Adversarial prior: As the per-episode probability of deception varies from 0.1 to 0.9, the optimal threshold under F2 (penalty) decreases monotonically from $\tau^* = 1.00$ to $\tau^* = 0.04$—a 25$\times$ change driven entirely by prior beliefs about the deployment environment (Fig. 4, center). In high-adversarial-rate regimes the optimizer prefers aggressive detection because false alarms on rare honest agents barely affect collision rates.

c) Oracle comparison: To bound achievable performance, we benchmark against a ground-truth oracle that perfectly knows the adversarial type from $t = 0$ and a naive agent with no detection. At $\sigma_{\text{obs}} = 2.0$, the naive agent collides in 4.0% of episodes, the F2-penalty detector collides in 1.5%, the oracle collides in 1.5% (Fig. 4, right). The detector recovers 100% of the preventable collision gap at this noise level: the residual 1.5% reflects scenarios in which even ground-truth knowledge is insufficient for B to brake in time.

d) Seed robustness: The F2 penalty method's optimum is invariant to base-seed choice across 10 seeds: $\tau^* = 0.6303 \pm 0.0000$, $\ell = 0.238 \pm 0.002$ s, recall $79.7 \pm 0.3\%$. Numbers reported elsewhere in this paper are not Monte Carlo artifacts.

D. Ablation: Conflict-Phase Gating

The detector's inconsistency score is gated to the conflict phase. Once the other agent has fully cleared the intersection, observations no longer update the score. This rule was introduced to prevent false alarms during legitimate post-yielding acceleration. An honest yielder that has stopped and is resuming has acceleration above the yield threshold while still signaling, which the score would otherwise treat as deception. To test whether the gate is necessary, we re-run the detector at five threshold values with the gate disabled (the detector keeps scoring for the entire episode) and compare against the standard gated variant. All other parameters are held fixed.

Disabling the gate appears at first glance to improve performance: recall rises to near 100% at every τ (Table II). This is misleading. The high recall reflects detections that fire *after* the adversary has already cleared the intersection, i.e. too late to be acted on. The collision rate is identical at every τ between the two variants, confirming that those late detections do not prevent any additional collisions.

At the same time, the gated variant achieves lower mean detection latency at every $\tau \geq 0.5$ (e.g., 0.207 s vs. 0.234 s at $\tau = 0.5$ and 0.401 s vs. 0.482 s at $\tau = 1.0$), because gating restricts the recall metric to detections that occurred during the conflict, where they could plausibly have informed action. The gate does not change *whether* adversarial behavior is eventually noticed, instead it defines what counts as an *actionable* detection. We treat the gate as part of the detector specification rather than a tunable parameter.

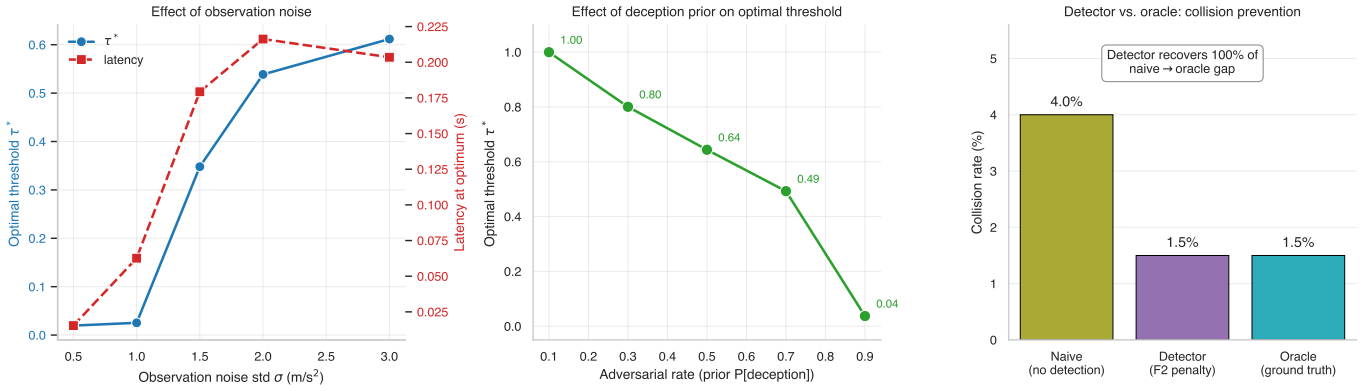


Fig. 4. Sensitivity analyses. Left: τ^* and latency at the optimum versus observation noise standard deviation. Center: τ^* under F2 penalty as a function of the adversarial-episode rate. Right: collision rates of naive, detector, and oracle agents at $\sigma_{\text{obs}} = 2.0$.

TABLE II
ABLATION: DETECTOR PERFORMANCE WITH AND WITHOUT CONFLICT-PHASE GATING, HOLDING τ AND ALL OTHER PARAMETERS FIXED.

τ	Gating	Collision (%)	Latency (s)	FA (%)	Recall (%)
0.1	on	1.5	0.113	16.5	84.9
0.1	off	1.5	0.105	16.5	100.0
0.3	on	1.5	0.152	13.5	83.0
0.3	off	1.5	0.163	13.5	100.0
0.5	on	1.5	0.207	8.5	81.1
0.5	off	1.5	0.234	8.5	100.0
0.7	on	2.0	0.272	5.0	77.4
0.7	off	2.0	0.323	5.0	99.1
1.0	on	2.5	0.401	3.0	69.8
1.0	off	2.5	0.482	3.0	96.2

VI. CONCLUSION

We have formulated and solved the problem of selecting a detection threshold τ for an active inference agent operating in an intersection with potentially deceptive yielding signals. A naive trusting agent collides on every deterministic encounter with a deceptive adversary. A detector that flags signal-kinematics inconsistencies eliminates this failure mode and, at observation noise $\sigma_{\text{obs}} = 2.0\,\text{m/s}^2$, matches a ground-truth oracle in collision prevention. The three optimization formulations produced complementary findings.

The scalarized formulation (F1) revealed that the optimal threshold spans a $20\times$ range across reasonable weight settings, indicating that threshold selection is genuinely a values-laden design decision rather than a technical one. The constrained formulation (F2), solved by both quadratic penalty and log barrier methods, showed the characteristic boundary-approach behavior of each: penalty crosses the constraint from the infeasible side and lands exactly on the boundary, while barrier remains strictly feasible and approaches asymptotically. At a fixed false alarm bound of 5%, F2-penalty achieved lower latency (0.245 s) than F1's nearest weighted-sum analog (0.316 s), supporting the case for explicit constraint formulation when a hard operational bound exists.

The multi-objective analysis (F3) validated both. Five of six optima from F1 and F2 land on the empirical Pareto frontier. Sensitivity analyses identified the regimes in which threshold optimization matters. Observation noise governs whether a trade-off exists at all: below $\sigma_{\text{obs}} = 1.0\,\text{m/s}^2$ the problem is degenerate and the optimum binds at the lower boundary. Above this, a smooth interior optimum emerges. The prior on adversarial encounters drives τ^* over a $25\times$ range, underscoring that the right threshold depends on the deployment environment.

A. Broader Implications

Beyond the specific intersection setting, this work illustrates a more general point about embodied AI systems that consume explicit communication channels. Vehicle-to-vehicle communication, gesture recognition, and intent-prediction modules all share the same structural risk: an input channel that is informative when honest and corrosive when adversarial. The standard response in the autonomy literature is to either trust the channel (and inherit its failure modes) or ignore it (and forgo coordination benefits). Our results suggest a middle path that is both principled and tractable: treat trust calibration as an explicit optimization problem with measurable performance bounds, rather than a hyperparameter chosen by intuition.

The fact that the optimal τ shifts by an order of magnitude with the deployment prior on adversaries means a single "universally robust" threshold does not exist. A deployer at an isolated rural intersection (low adversarial rate) should configure detection differently than a deployer in a contested urban environment, and our framework makes that configuration choice explicit and defensible. More broadly, the observation that constrained formulation outperforms weighted-sum proxies at fixed false alarm bounds is a useful lever for autonomy practitioners: hard operational requirements (e.g., a regulator-imposed FA ceiling) are more faithfully encoded as constraints than as terms in a scalar loss.

B. Limitations and Future Work

The model abstracts several aspects of real intersection interactions. The agent's motion is one-dimensional and the action set discrete—both could be relaxed without changing the optimization structure. The adversarial policy is a single behavior (constant velocity with signal) rather than a distribution over adversarial strategies, which limits the strength of robustness claims. Extending to a parametric family of adversaries, like varying signal-onset timing, partial deceleration, intermittent signaling, would yield a worst-case robustness analysis and is a natural extension.

The detector itself is a single scalar threshold on a hand-designed score. Learned detectors could plausibly improve the latency–false alarm trade-off, e.g. by exploiting richer features than our smoothed inconsistency score captures or by adapting to the deployment distribution rather than relying on hand-tuned parameters like a_{yield} and $v_{\min}$. Though, this would be at the cost of the interpretability that motivated this formulation and at the cost of introducing the very trust calibration problem that motivated the threshold-based approach. A learned score has no closed-form relationship to the quantities a deployer or regulator might want to bound, and its outputs become a new channel whose reliability under adversarial inputs would itself need to be characterized.

Finally, the EFE objective omits the entropy term that gives active inference its name on the grounds that passive observation makes it nearly invariant to action. Restoring information-gathering as a real choice (for example, by giving the agent its own signaling action or letting it actively probe by varying its velocity) would re-engage the full canonical formulation. We expect the broader research direction—threshold optimization for adversarial robustness in communicating multi-agent systems—to generalize beyond traffic to settings such as drone coordination, robot fleet management, and multi-agent reinforcement learning with explicit communication channels.

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